New Data Provenance Requirement, Need to Capture Decision-Making Data! Lessons from an autonomous neutron powder diffraction experiment

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What we did

In May 2025, we ran a 24-hour autonomous experiment on the Nanoscale-Ordered Materials Diffractometer (NOMAD; https://neutrons-user-facilities.ornl.gov/instruments/nomad/) at the Spallation Neutron Source (SNS) to explore the Morin magnetic transition in hematite (α-Fe2O3, near 260 K). The Interconnected Science Ecosystem (INTERSECT; https://www.ornl.gov/intersect/) linked the sample temperature control to analysis and decision-making services, which chose each next temperature to measure.

What went “wrong”: the data survived, the reasons did not

The facility did its job well. It permanently stores the raw neutron event files, reduced diffraction patterns and run metadata (run number, title, start time, temperature logs, sample) for all 600 runs, and its public catalog, ONCat (https://oncat.ornl.gov/), makes them findable, accessible, interoperable and reusable. But in one of the facility’s first autonomous experiments, we did not save:

- the campaign definition: which services ran, with which settings (we save this now)
- the messages between services, such as “measure at 240 K, before the transition” (still not saved)

We rebuilt which runs were autonomous from run titles, times and notes; Auditing post-experiment takes lots of effort.

What we fixed by hand

After the experiment, we sorted the runs into 13 groups (temperature ramps, tests, debugging, autonomous runs), fixed metadata entered wrongly during debugging, and separated the successful autonomous campaigns from the failed starts mixed in with them. These fixes are plain text tables and scripts that anyone can check, but they live in our project repository, not with the facility record, so the next user will still find the original errors.

What a release has to ship with

For autonomous experiments, the data alone is not enough. A release should also include:

1. **The campaign definition**. Saved for every new experiment; tells which services ran and able to re-run
2. **A record of the decisions.** This is the part we still do not keep. A 24-hour period can produce 1,000 to 100,000 messages, and some experiments run for days. We need either a place to store logs at that scale or agreement on what is enough to keep, for example only the decision messages
3. **Corrections next to the original metadata.** Need to stay with facility, on-going work with SNS

Why this matters

In an autonomous experiment, the "method" is a set of services written by different people and teams. The campaign definition names each service and its version. It can double as the list of software to cite. Without it, the authors of the services that made the decisions are invisible, even though their code chose every data point. Saving the campaign definition with the data is the simplest way to give them credit.

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